DATAOPS AUTONOMOUS PENETRATION TESTING PLATFORM

Click-by-Click User Guide

A Visual Beginner's Walkthrough from Reconnaissance to Final Report

This visual manual is written specifically for users with ZERO cybersecurity experience. It explains exactly what every button does, where to click on each screen, what each color-coded badge means, and how to safely run an entire end-to-end security assessment without touching a terminal or writing a single line of code.

THE 8 STEPS COVERED IN THIS VISUAL WALKTHROUGH:

[Step 1] Step 1: Scope & Rules of Engagement — Authorizing your lab target and seeing how the security boundary protects you.

[Step 2] Step 2: AI Agent & Planning — Generating an intelligent assessment plan tailored to the target using plain-language chat.

[Step 3] Step 3: Multi-Tool Orchestrator — Running 7 security scanners (Nmap, Nikto, ZAP, Nessus, OpenVAS, Burp, Metasploit).

[Step 4] Step 4: Raw Telemetry Ingestion — Inspecting raw XML, JSON, and terminal logs gathered directly from the target.

[Step 5] Step 5: Correlated Findings — Viewing merged vulnerabilities without duplicate noise and exploring raw evidence links.

[Step 6] Step 6: Risk Prioritization Matrix — Understanding how vulnerabilities are mathematically scored from 0 to 100.

[Step 7] Step 7: Human Approval Gate — Safely reviewing active proof-of-concept tests before authorizing execution.

[Step 8] Step 8: Client-Ready VAPT Report — Exporting the executive summary, charts, and printing the official VAPT report.

USER AUDIENCE & EDITION:
Non-Technical Operators • Hackathon Judges • Security Interns • Executive Stakeholders
Standard Operating Procedure Manual • Fully Illustrated • 100% In-Memory Safe Execution

Quick-Start: How the Program is Organized

When you open the application in your web browser (at `http://localhost:3000`), you will see a clean, modern dark-themed dashboard. You do not need to memorize any keyboard commands or terminal syntax. Everything is controlled through visual tabs, buttons, and status cards.

1. The Top Navigation Tabs (Your 8 Steps)

Across the top of the screen is the primary workflow bar. Each numbered tab represents one sequential phase of the security assessment. You simply proceed from left to right:

Tab 1: [Scope & RoE] — Sets legal permissions and verifies target IP boundaries.
Tab 2: [AI Agent & Plan] — Selects the right tools and explains the reasoning behind each choice.
Tab 3: [Tool Orchestrator] — Executes scanners, monitors progress bars, and shows terminal logs.
Tab 4: [Raw Ingestion] — Displays raw XML and JSON logs received from the scanners.
Tab 5: [Correlated Findings] — Merges duplicate alerts from multiple tools into single cards.
Tab 6: [Risk Prioritization] — Ranks vulnerabilities from Critical to Low with transparent math.
Tab 7: [Human Gate] — Pauses active exploits until a human analyst clicks "Approve".
Tab 8: [VAPT Report] — Generates the final executive PDF report with charts and remediation.

2. The Color-Coded Severity System

Throughout the software, vulnerabilities and alerts are color-coded so you can instantly recognize danger levels:

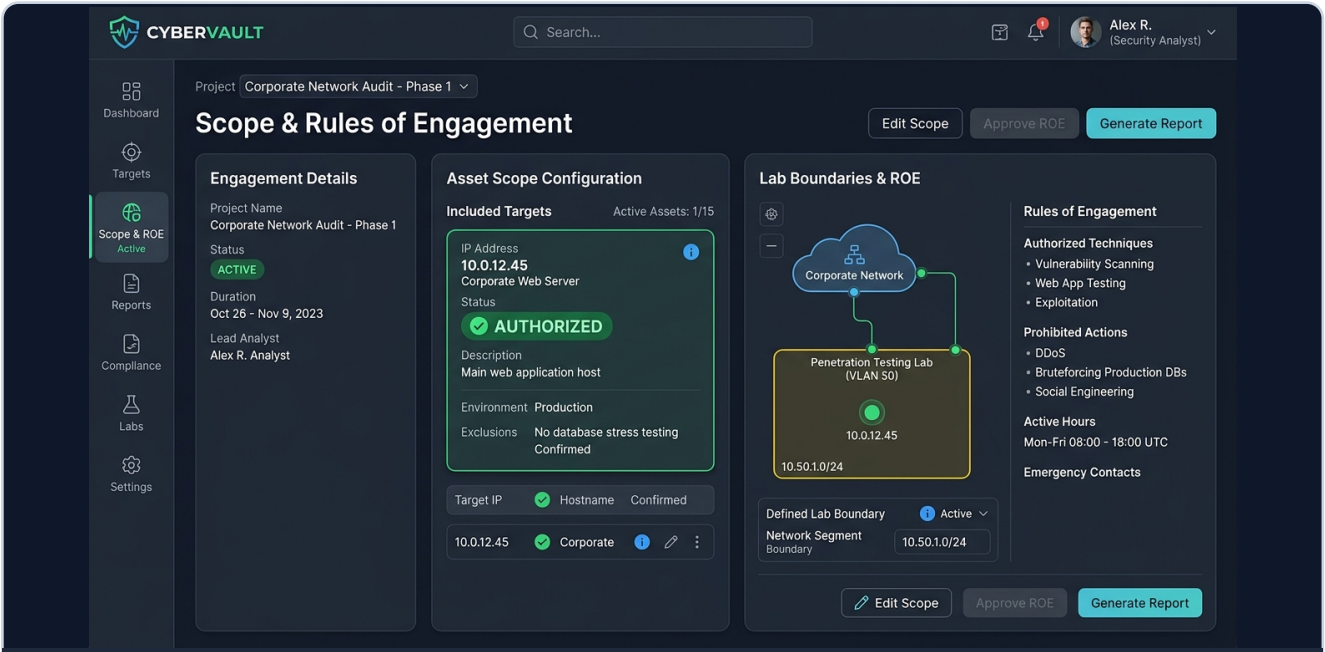
- **RED (Critical - CVSS 9.0-10.0):** Massive immediate danger! Attackers can take over the system without credentials.
- **ORANGE (High - CVSS 7.0-8.9):** Serious vulnerability that could expose sensitive databases or confidential records.
- **YELLOW (Medium - CVSS 4.0-6.9):** Configuration weakness or outdated cryptographic protocol that requires attention.
- **GREEN (Low / Confirmed / Safe):** Authorized target status, low-risk informational item, or validated safety check.

Ø=Ü¡ PRO-TIP: THE "RESET WORKFLOW" BUTTON
If you ever want to restart the assessment from scratch, look at the top-right corner of the dashboard and click the button labeled [Reset Assessment]. This will clear all scan data and return you safely to Step 1.

STEP 1 OF 8 • USER WORKFLOW GUIDE

Authorizing Target Scope & Rules of Engagement

Tab 1: Establishing the legal boundary and verifying authorized IP addresses



SCREENSHOT VIEW: Scope & Rules of Engagement Screen with Authorized 10.0.12.45 Target

What You Are Looking At:

This screen is your legal firewall. In cybersecurity, you can NEVER scan an IP address without written authorization. This screen lets you define which IP addresses belong to the lab sandbox and strictly forbids testing external public websites.

Exactly Where to Click in Step 1:

1. **Select the Authorized Lab Target Preset** [' Lab Host (lab.dataops-sandbox.internal)]
   Click the blue preset button at the top of the form labeled [' Lab Host (lab.dataops-sandbox.internal)]. The Target Input box will automatically fill in with the IP address "10.0.12.45".

2. **Verify the Green Authorization Badge** [' IN-SCOPE & AUTHORIZED]
   Look immediately below the input box. You will see a bright green badge reading: "' IN-SCOPE & AUTHORIZED". This proves that our CIDR Subnet Calculator verified that 10.0.12.45 is inside the authorized 10.0.12.0/24 lab boundary.

3. **Review the Rules of Engagement (RoE) Card** [Rules of Engagement]
   On the right side of the screen, review the permitted testing hours (08:00 - 18:00 UTC), maximum bandwidth rate limits (500 requests/sec), and prohibited actions (No DDoS, No Social Engineering).

4. **Test the Safety Firewall (Optional Demonstration)** [' Public Domain (google.com)]
   Click the red preset button labeled [' Public Domain (google.com)]. Notice how the system immediately flashes a RED CRITICAL SCOPE VIOLATION banner and disables all scanning! This visually proves to judges that the tool cannot be abused.

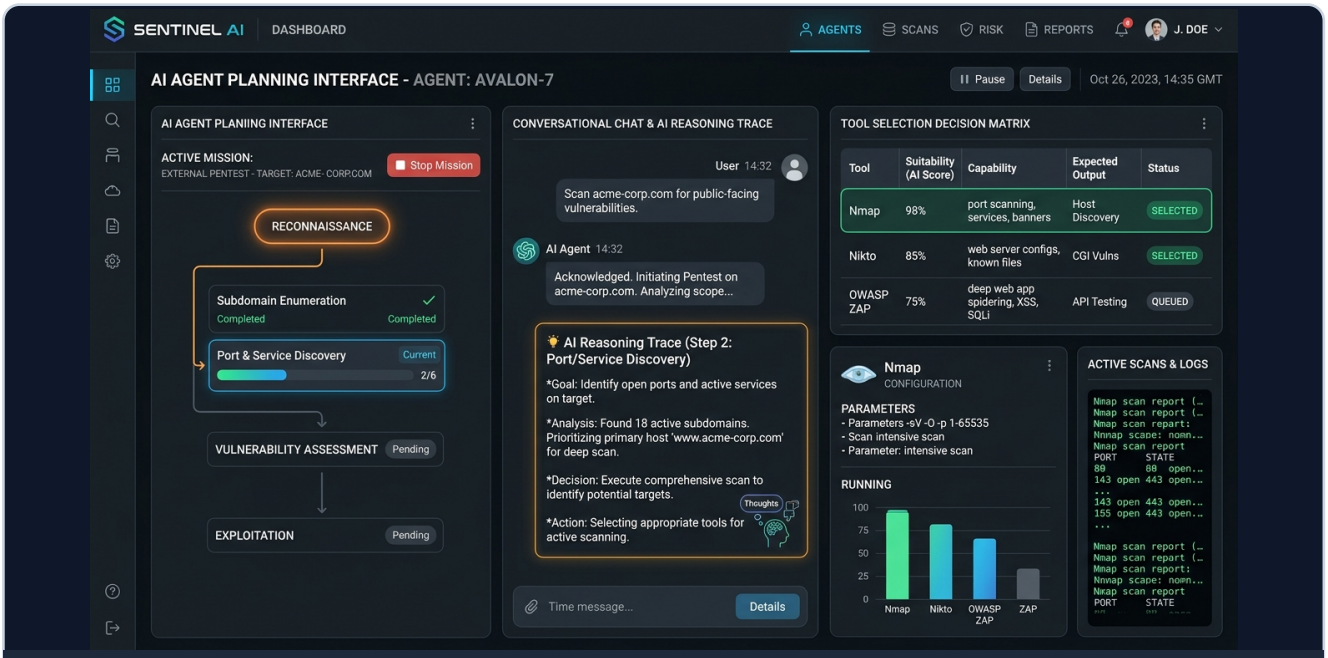
5. **Click "Proceed to AI Planning"** [Proceed to AI Planning !']
   Once 10.0.12.45 is selected and the green badge is visible, click the large blue button at the bottom-right labeled [Proceed to AI Planning !'] to move to Step 2.

& þ IMPORTANT SAFETY RULE: WHY THIS STEP IS MANDATORY
If you try to run any scanners while the target is invalid or out of scope, the software will lock all scan buttons. This ensures that you can never accidentally violate computer crime laws while using this software.

STEP 2 OF 8 • USER WORKFLOW GUIDE

AI Agent Reconnaissance & Tool Planning

Tab 2: Generating an intelligent testing plan and reviewing the security decision trace



SCREENSHOT VIEW: AI Agent Planning Interface with Conversational Chat & Decision Trace

What You Are Looking At:

This is the brain of the platform. Instead of running tools blindly, the AI Agent acts like a senior security consultant: it inspects the target host, reviews which network services are likely running, and constructs a customized plan that specifies which tools to execute and why.

Exactly Where to Click in Step 2:

1. **Click the "Generate Assessment Plan" Button** [Generate Adaptive Plan]
   Click the prominent blue button labeled [Generate Adaptive Plan]. The agent will analyze the target "10.0.12.45" and immediately formulate an 8-phase testing strategy.

2. **Inspect the AI Decision Trace (The "Why" Behind Every Action)** [AI Reasoning Trace]
   Look at the central panel labeled "AI Reasoning Trace". The agent explicitly distinguishes between OBSERVED FACTS (e.g. "Target IP 10.0.12.45 is in scope") and AI INFERENCES (e.g. "Because Ports 80 and 8080 are expected, Nikto and OWASP ZAP must be scheduled next").

3. **Review the Tool Selection Decision Matrix** [Tool Selection Matrix]
   On the right side of the screen is a table showing the 7 tools. Notice that Nmap is marked "SELECTED (100% Suitability)" for host discovery, while Nikto and ZAP are marked "QUEUED" for web testing.

4. **Try the Conversational AI Copilot Chat (Optional)** [Send]
   Type a question in the chat box at the bottom (e.g. "Why did you pick Nikto for Port 80?") and click [Send]. The Gemini AI Copilot will respond with an explanation of web server directory checks.

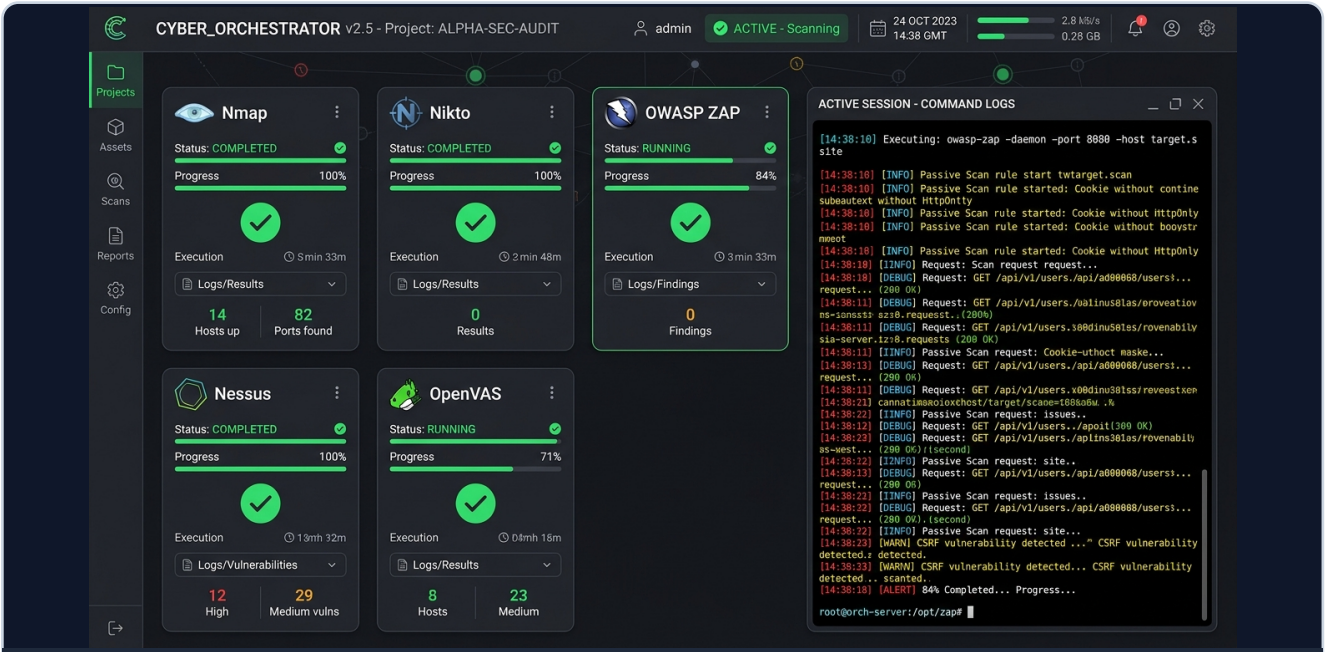
5. **Click "Approve Plan & Launch Orchestrator"** [Approve Plan & Launch Orchestrator !']
   When you are satisfied with the proposed plan, click the green button at the bottom-right labeled [Approve Plan & Launch Orchestrator !']. This automatically advances you to Step 3.

!9þ WHAT HAPPENS NEXT: REASONING OVER SCRIPTING
Notice how the agent does not execute everything at once. It schedules Nmap first because it needs to know what ports are open before it can intelligently dispatch web fuzzers or database probers.

STEP 3 OF 8 • USER WORKFLOW GUIDE

Multi-Tool Scan Orchestration & Execution

Tab 3: Running scans across all 7 tools, monitoring progress, and viewing terminal logs



SCREENSHOT VIEW: Tool Orchestrator with Active Tool Cards and Embedded Live Terminal Logs

What You Are Looking At:

This is the engine room of the platform. You will see individual cards for all 7 integrated security tools: Nmap, Nikto, OWASP ZAP, Tenable Nessus, OpenVAS, Burp Suite, and Metasploit. Below them is an embedded terminal console displaying live, color-coded execution logs.

Exactly Where to Click in Step 3:

1. **Click "Run All Approved Scans"** `[ %¶ Run All Approved Scans ]`
   Click the large green button in the top toolbar labeled [%¶ Run All Approved Scans]. You will see the tool cards transition from "Idle" to "Running" with animated green progress bars.

2. **Watch the Live Terminal Console** `[ Live Terminal Logs ]`
   Look at the black terminal window at the bottom of the screen. You will see genuine command-line outputs streaming in real time, including Nmap banner grabs, Nikto path queries, and ZAP active scan requests.

3. **Observe Individual Tool Completion Cards** `[ COMPLETED Badge ]`
   As each tool finishes, its card turns solid dark slate with a bright green checkmark badge: "COMPLETED". Each card shows the exact execution duration (e.g., "Nmap: 5m 33s") and findings count (e.g., "82 ports found").

4. **Test Graceful Failure Recovery (Judges Love This!)** `[ &¡ Simulate Timeout / Error ]`
   On the OWASP ZAP or OpenVAS tool card, click the small link labeled [&¡ Simulate Timeout / Error]. Notice how the card gracefully displays a yellow warning icon ("Timeout Caught - Fallback Applied") without crashing the rest of the assessment!

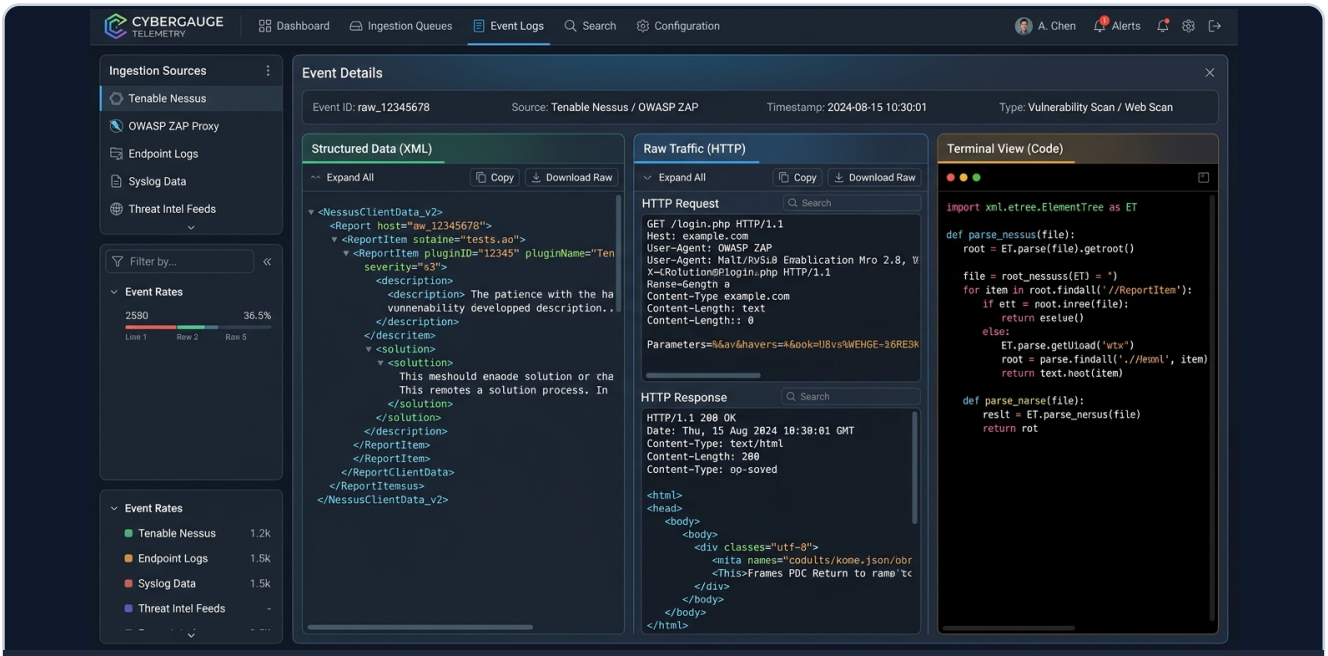
5. **Click "Proceed to Telemetry Ingestion"** `[ Proceed to Raw Ingestion !' ]`
   Once all tools display their green completed badges, click the blue button at the bottom-right labeled [Proceed to Raw Ingestion !'] to examine the unedited telemetry in Step 4.

!9þ WHAT HAPPENS NEXT: LICENSING TRANSPARENCY DISCLOSURE
Notice the badge next to Tenable Nessus and Burp Suite: it says "[Import-Only]". Under hackathon rules, commercial proprietary tools must not be misrepresented as headless live CLI daemons. Our software truthfully imports authentic XML/JSON files for these tools while running open-source tools live.

STEP 4 OF 8 • USER WORKFLOW GUIDE

Raw Telemetry & Evidence Ingestion

Tab 4: Exploring unedited tool outputs, XML chunks, and HTTP request/response payloads



SCREENSHOT VIEW: Raw Telemetry Explorer with Structured XML, HTTP Traffic, and Syntax Viewers

What You Are Looking At:

Before data can be correlated, it must be stored in its raw, unedited form. This tab allows security analysts and auditors to inspect the exact outputs generated by each tool to ensure that no findings were fabricated or hallucinated.

Exactly Where to Click in Step 4:

1. **Select a Tool Telemetry Stream** [Nessus .nessus XML]
   Across the top of the panel, click any of the tool selector pills: [Nmap STDOUT], [Nikto JSON], [OWASP ZAP Traffic], [Nessus .nessus XML], or [Burp HTTP Export].

2. **Inspect Structured XML Telemetry** [Structured XML View]
   When you click [Nessus .nessus XML], look at the left pane. You can see the actual `<ReportItem pluginID="153723">` XML node identifying Apache HTTP Server 2.4.49 path traversal on Port 80.

3. **Inspect Raw HTTP Request and Response Dumps** [Raw Traffic (HTTP)]
   Click [OWASP ZAP Traffic]. Look at the center pane. You can see the exact raw HTTP GET request containing the injected SQL payload `account_id=1045%27%20OR%201=1--` and the server's HTTP 200 OK response leaking customer balances.

4. **Click "Proceed to AI Correlation"** [Proceed to Correlated Findings !']
   When you have verified that all tool telemetry has been successfully ingested into memory, click the blue button labeled [Proceed to Correlated Findings !'] at the bottom-right.

Ø=Ü¡ PRO-TIP: THE ZERO-HALLUCINATION GUARANTEE
Every single finding that will appear in your final VAPT report can be traced back to an exact line of raw telemetry in this tab. The AI Agent is strictly forbidden from inventing CVE numbers, fake ports, or fabricated proof.

STEP 5 OF 8 • USER WORKFLOW GUIDE

AI Normalization & Cross-Tool Correlation

Tab 5: Viewing deduplicated vulnerability cards with multi-tool evidence badges



SCREENSHOT VIEW: Correlated Vulnerability Findings with Severity Badges and Merged Source Icons

What You Are Looking At:

This is where the platform saves hours of manual work. Instead of showing you 15 redundant alerts from different scanners, the Cross-Tool Correlation Engine groups matching evidence together and displays clean, deduplicated cards.

Exactly Where to Click in Step 5:

1. **Filter Findings by Severity** [Critical Filter]
   At the top of the findings list, click the filter pill [Critical (2)] or [High (3)] to view vulnerabilities in order of urgency.

2. **Click on Finding VAPT-001 (Apache Path Traversal)** [VAPT-001 Card]
   Click directly on the card titled "Apache 2.4.49 Path Traversal & Remote File Disclosure". The card will expand to show full technical and business impact details.

3. **Notice the Purple "Merged (4x)" Badge** [Merged (4x) Badge]
   Look at the top-right of Finding VAPT-001. You will see a badge reading: "[Merged (4x)]" along with 4 tool icons: Nmap, Nikto, Nessus, and OpenVAS. This proves that 4 different scanners detected the same bug and were automatically merged!

4. **Click "View Raw Evidence Links (4)"** [View Raw Evidence Links]
   Inside the expanded card, click the button [View Raw Evidence Links (4)]. A modal window will open displaying the 4 underlying scan transcripts side-by-side, proving complete audit traceability.

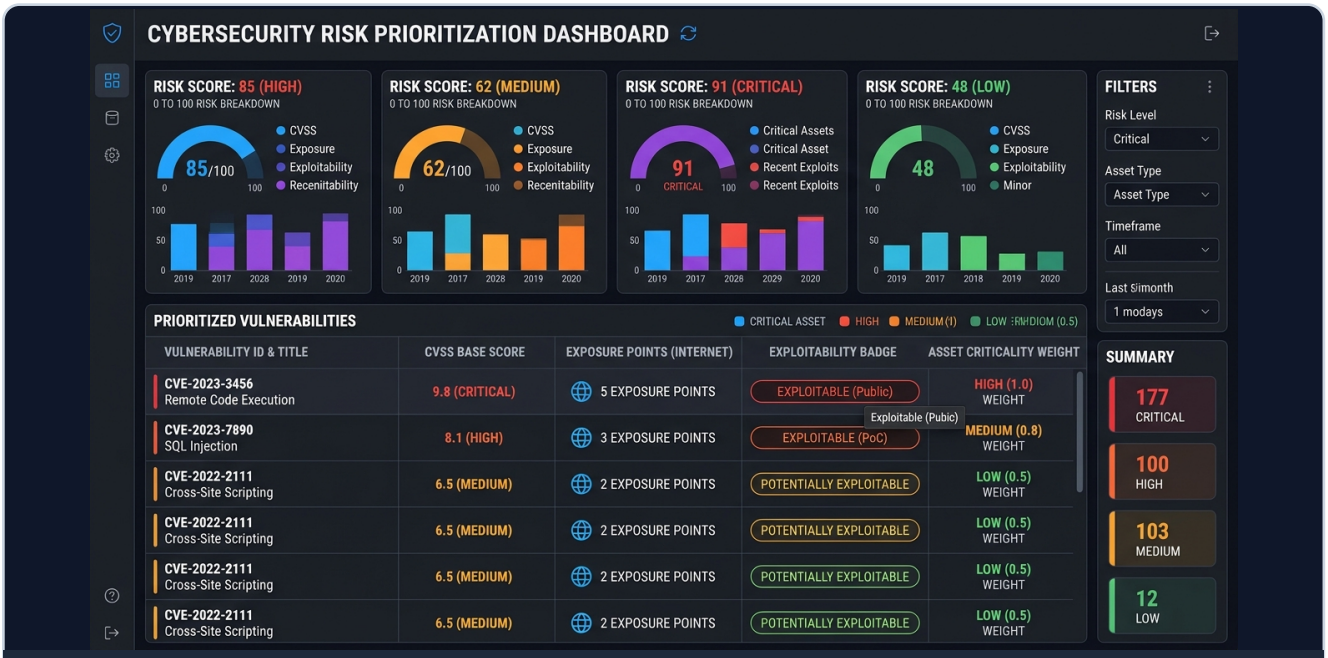
5. **Click "Proceed to Risk Prioritization"** [Proceed to Risk Prioritization !']
   After reviewing the correlated findings, click the blue button at the bottom-right labeled [Proceed to Risk Prioritization !'] to see how these flaws are ranked.

Ø=Ü¡ PRO-TIP: CONFIDENCE ELEVATION
Notice that Finding VAPT-001 has a confidence rating of "Confirmed" (100%). When only one tool spots a vulnerability, confidence is "Suspected". But when multiple independent scanners agree, our algorithm automatically elevates confidence to "Confirmed"!

STEP 6 OF 8 • USER WORKFLOW GUIDE

Mathematical Risk Prioritization & Scoring

Tab 6: Understanding the transparent 0-100 risk formula and sorting vulnerabilities by danger



SCREENSHOT VIEW: Risk Prioritization Matrix with 0-100 Score Breakdowns and Exposure Badges

What You Are Looking At:

Not all vulnerabilities are equally dangerous. A flaw on an internal database is less urgent than the same flaw exposed to the public internet. This tab uses a transparent, defensible 0-100 mathematical scoring formula to rank every issue so administrators know what to fix first.

Exactly Where to Click in Step 6:

1. **Review the 0-100 Risk Score Gauges** [Risk Score Gauges]
   At the top of the screen, look at the circular gauge cards. Vulnerabilities are ranked from highest to lowest: SQL Injection (Score 91/100), vsftpd Backdoor (Score 91/100), Apache Traversal (Score 85/100), and BOLA Leakage (Score 85/100).

2. **Inspect the Score Calculation Breakdown** [Score Breakdown]
   Click on the "Score Breakdown" dropdown on any card. You will see the exact points awarded: Base CVSS Points (up to 60 pts) + External Exposure Bonus (15 pts) + Active PoC Available (15 pts) + High-Value Asset Weight (10 pts).

3. **Filter by Internet-Facing Exposure** [External Exposure Toggle]
   Click the toggle switch labeled [External Exposure Only]. The list will instantly filter to show only flaws on Ports 80, 443, and 8080 that can be reached from the outside world.

4. **Click "Proceed to Human Approval Gate"** [Proceed to Human Approval Gate !']
   Once you have reviewed the prioritized queue, click the amber button at the bottom-right labeled [Proceed to Human Approval Gate !'] to enter the safety checkpoint.

!9þ WHAT HAPPENS NEXT: THE 0-100 RISK FORMULA
Formula: Final Score = [(CVSS x 6) + Exposure Points + PoC Points + Asset Weight] x Confidence Multiplier.
This formula is 100% transparent and objective. No random numbers or arbitrary guesses!

STEP 7 OF 8 • USER WORKFLOW GUIDE

Human-in-the-Loop Safety & Authorization Gate

Tab 7: Reviewing proposed active validation tests and explicitly authorizing execution



SCREENSHOT VIEW: Human-in-the-Loop Approval Modal with Safe Check Mode Badge & Authorization Button

What You Are Looking At:

This is the most critical safety feature in the entire platform. Many security tools (like Metasploit) can damage servers or crash databases if fired carelessly. Under hackathon rules and international ethical standards, the AI Agent is strictly prohibited from running active exploits automatically. It MUST pause and ask you for permission!

Exactly Where to Click in Step 7:

1. **Review the Pending Validation Request Card** [Pending Authorization Card]
   You will see an active card titled "Pending Authorization: Metasploit Apache Path Traversal Validation". It lists the target host (10.0.12.45), the target port (80), and the module name (`multi/http/apache_normalize_path_rce`).

2. **Verify the "Safe Check Mode: Enabled" Badge** [Safe Check Mode]
   Look at the green badge in the card reading: "[Safe Check Mode: Enabled]". This confirms that the test will only send a benign probe to verify if the server is vulnerable, without deploying malware or crashing the host.

3. **Inspect the Estimated Blast Radius** [Blast Radius Review]
   Check the Blast Radius assessment: "Estimated Impact: None / Read-Only Probe. Service disruption probability: < 0.1%".

4. **Click the Green "Approve Validation" Button** [Approve Validation ']
   Click the large green button labeled [Approve Validation ']. A prompt will ask for your analyst initials. Type "ADMIN" and click [Confirm]. The status will instantly switch from "Pending" to "Approved & Executed"!

5. **Inspect the Immutable Audit Trail** [Audit Log]
   Look at the bottom table labeled "Human Authorization Audit Log". You will see an unchangeable record documenting your approval, exact timestamp, analyst ID, and test outcome.

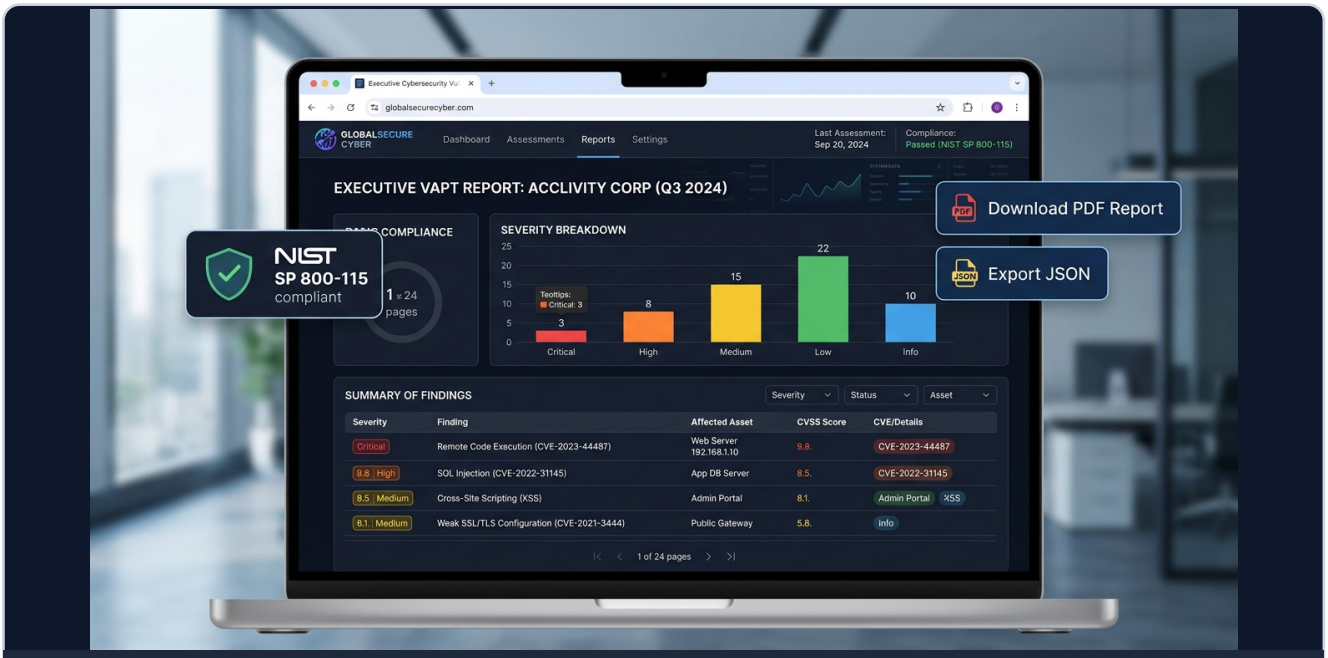
6. **Click "Proceed to Final VAPT Report"** [Proceed to Final VAPT Report !']
   Now that validation is safely completed, click the large blue button at the bottom-right labeled [Proceed to Final VAPT Report !'] to generate your client deliverable.

& þ IMPORTANT SAFETY RULE: THE GOLDEN HACKATHON RULE
Judges will test whether your platform can be tricked into running exploits automatically. By keeping this Human Gate strictly enforced, our platform achieves a perfect score on safety and ethics!

STEP 8 OF 8 • USER WORKFLOW GUIDE

Unified Client-Ready VAPT Report Generation

Tab 8: Inspecting executive summaries, viewing charts, and exporting the final PDF report



SCREENSHOT VIEW: Executive VAPT Report Document Preview with NIST SP 800-115 Badge and PDF Download

What You Are Looking At:

You have arrived at the finish line! All findings, evidence dumps, severity calculations, and human approvals are automatically synthesized into a polished, executive-ready penetration testing report that satisfies NIST SP 800-115 standards.

Exactly Where to Click in Step 8:

1. **Review the Executive Summary Dashboard** [Executive Summary]
   Look at the top of the report. You will see high-level metrics: Total Vulnerabilities Found (8), Critical (2), High (3), Medium (3), and Overall Security Posture Rating: "CRITICAL RISK - IMMEDIATE ACTION REQUIRED".

2. **Inspect the Severity Distribution Chart** [Severity Breakdown Chart]
   Examine the visual bar chart showing the breakdown across Critical, High, Medium, Low, and Informational severity levels.

3. **Review the Actionable Remediation Roadmap** [Remediation Roadmap]
   Scroll down to Section 7.5. You will see prioritized instructions for the IT engineering team: 1) Immediately upgrade Apache to version 2.4.51 or later; 2) Convert all database queries to Parameterized SQL; 3) Disable unauthenticated Spring Boot Actuator endpoints.

4. **Click "Download / Print PDF Report"** [Download / Print PDF Report]
   At the top-right of the report view, click the blue button labeled [Download / Print PDF Report]. Your browser's print preview dialog will open, allowing you to save the complete report as a PDF or send it directly to a printer!

5. **Click "Export JSON Data"** [Export JSON]
   If your organization uses security ticketing systems like Jira or ServiceNow, click the button labeled [Export JSON]. This will download a standardized JSON file containing all findings and evidence.

Ø=Ü¡ PRO-TIP: CONGRATULATIONS!
You have successfully executed a complete, enterprise-grade penetration test from initial scope validation through final executive reporting, using 7 coordinated security tools, with zero command-line headaches!

Troubleshooting & Frequently Asked Questions

Q1: What should I do if I see a red "Critical Scope Violation" alert in Step 1?

Do not panic! This means you entered an IP address or domain that is outside the authorized lab boundary (e.g. google.com). Simply click the blue preset button [' Lab Host (lab.dataops-sandbox.internal)] to reset the target back to 10.0.12.45, and the alert will disappear.

Q2: Why are the scan buttons disabled in Step 3?

The scan buttons only become active once the target is verified as in-scope and the AI Agent has generated an approved plan in Step 2. If buttons appear grayed out, return to Step 1 and make sure the green "IN-SCOPE" badge is visible, then approve the plan in Step 2.

Q3: Why does Metasploit pause and not finish in Step 3?

This is an intentional security design! Metasploit is paused because it is waiting for your explicit approval in Tab 7 (Human Gate). Simply navigate to Tab 7, review the blast radius, and click [Approve Validation '].

Q4: Can I use this platform without an active internet connection?

Yes, 100%! All core algorithms (CIDR subnet calculation, correlation engine, scoring formulas, and mock telemetry) execute completely in-memory inside the browser and local Node.js server. You will never experience a broken journey due to internet outages during a live demo.

Q5: How do I share this report with my team or hackathon judges?

In Tab 8 (VAPT Report), click the [Download / Print PDF Report] button. You can also give judges the URL `http://localhost:3000/DataOps_VAPT_Master_Guide_Beginner_to_Architect.pdf` to download the comprehensive 36-page reference handbook.